# ANDRÉ NICOLAS SILVA

**Full Stack Engineer | Product, AI, Systems & Software Engineering**

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**Portfolio** ▪ GitHub ▪ LinkedIn

## PROFESSIONAL SUMMARY

Full Stack Engineer with a **product and problem-solving focus**, experienced in building and evolving production SaaS products used by **10+ companies**. Works across product requirements, technical decisions, architecture, frontend, backend, APIs, data, AI, cloud, debugging, and production delivery.

Combines software engineering with an **AI-native development workflow**, using AI to accelerate implementation, explore solutions, understand codebases, review changes, debug issues, generate tests, and iterate on product functionality while maintaining technical judgment, architecture, correctness, and maintainability.

Strong experience with **TypeScript, Angular, React, Golang, REST APIs, SQL, PostgreSQL, Cloudflare, and Generative AI**, with the ability to move across application layers and translate **business problems and product requirements into working software**.

Experienced in taking ownership of problems rather than only individual tasks, balancing **product requirements, UX, business rules, technical constraints, system reliability, performance, and delivery**.

## CORE COMPETENCIES

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| **Product & Problem Solving** | Product Thinking, problem decomposition, requirements analysis, business rules, UX, feature ownership, scope definition, technical decision-making, independent delivery |
| **AI-Native Engineering** | Generative AI, LLM integration, prompt engineering, contextual prompting, structured AI responses, AI-assisted problem solving, AI-assisted coding, AI-driven development workflows |
| **Software Engineering** | Full-stack development, system design, software architecture, debugging, code review, automated testing, maintainability, reliability, production support |
| **Architecture & Systems** | Software Architecture, Hexagonal Architecture, Component-Based Architecture, frontend architecture, separation of concerns, reusable systems, distributed functionality |
| **Backend & APIs** | Golang, Node.js, Java, Spring/Spring Boot, REST APIs, internal APIs, backend services, concurrency, asynchronous processing, event-driven systems, Server-Sent Events (SSE), service integrations, automations |
| **Frontend & Web** | TypeScript, JavaScript, Java (MSPA and academic Android/SQLite), Node.js (MSPA and project work), Angular, React, Next.js, Angular Signals, RxJS, Web Components, Lit, Shadow DOM, Tailwind CSS |
| **Data** | SQL, PostgreSQL, Supabase, relational databases, database migrations, data persistence, queries |
| **Security & Integrity** | Input validation and sanitization, authorization and access control, privacy-aware consent flows, audit events, secure product safeguards |
| **AI Product Engineering** | Genkit, LLM-powered features, input sanitization, contextual information, structured outputs, reliable AI-powered product functionality |
| **AI Development Stack** | Codex, Claude, Cursor, CodeRabbit, Skills, MCPs, Agents, SDD, AI harnesses, deterministic evaluation, scope, output-quality, and model-cost guardrails |
| **Cloud & Infrastructure** | Cloudflare Workers, KV, D1, Durable Objects, AWS/Azure cloud services, Podman/Docker-compatible containers, Bash, Testcontainers, production Docker images, CI/CD, Git |
| **Performance & Quality** | Performance optimization, SEO, Lighthouse, Vitest, automated testing, code review, debugging, production reliability |

## PROFESSIONAL EXPERIENCE

**MSPA — Frontend Engineer (full-stack contributions)** *Apr 2025 – Present*

*SaaS platform for Privacy, Compliance (LGPD), and Digital Auditing*

- **Product Engineering & Problem Ownership:** Built and evolved core SaaS products, including Compass and Trust Center, used by **10+ companies**, taking ownership of problems from requirements and product context through technical definition, implementation, refinement, debugging, and production delivery.
- **Product & Technical Decision-Making:** Translated business requirements and product needs into technical solutions, contributing to feature scope, business rules, UX, architecture, implementation strategy, and trade-offs across application layers.
- **AI-Native Engineering:** Applied an AI-driven development workflow using **Codex, Claude, Cursor, CodeRabbit, Skills, MCPs, Agents, and SDD** to accelerate implementation, explore solutions, understand existing code, debug issues, review changes, and iterate on product functionality while maintaining architectural consistency and code quality.
- **Generative AI Product Engineering:** Integrated **Generative AI into production SaaS products**, developing user-facing capabilities with prompt engineering, contextual information, input sanitization, and structured responses using **Genkit**, focusing on reliable and consistent AI-powered functionality.
- **Full-Stack Engineering:** Delivered functionality across **frontend applications, backend services, APIs, data, and infrastructure**, moving between application layers according to the problem rather than being limited to a single technical layer.
- **Expanded Platform Stack:** Contributed at MSPA with **Node.js, Java, Spring/Spring Boot, AWS/Azure cloud services, Podman/Docker-compatible containers, Bash automation, Testcontainers, production Docker images, testing, accessibility, technical documentation, UI delivery, and security-oriented engineering**, selecting the appropriate layer according to product requirements.
- **Backend & Systems Engineering:** Worked with **Golang services, REST/internal APIs, concurrency, asynchronous processing, events, SSE, automations, and service integrations**, investigating and implementing solutions for production systems.
- **Data & Business Logic:** Worked with **SQL and PostgreSQL**, including queries, data persistence, migrations, business rules, and investigation of data-related issues across application workflows.
- **Software Architecture:** Participated in architectural decisions and implemented solutions involving **Hexagonal Architecture**, component-based architecture, separation of responsibilities, reusable systems, and maintainable application boundaries.
- **Web & Edge Systems:** Developed an agnostic consent engine using **Web Components, Lit, and Shadow DOM**, integrated with **Cloudflare Workers, KV, D1, and Durable Objects** for edge-based execution and distributed functionality.
- **Performance & Reliability:** Implemented performance and SEO improvements, investigating bottlenecks and improving loading behavior, usability, Lighthouse performance, and production reliability.
- **Engineering Coordination:** Reorganized development workflows during a critical project phase, improving scope clarity, sprint organization, and productivity across a **6-person team**, with direct coordination of 4 team members.

## OPEN SOURCE CONTRIBUTIONS

**Stoat / Revolt Platforms LTD — Volunteer Frontend Engineer** *2026 – Present*

Contribute to the open-source project with a focus on **UX engineering and frontend development**, identifying user-facing problems and turning them into technical improvements that make the product clearer and more intuitive. **Pull request approved by project maintainers.**

## PROJECTS

**Terto Beats — Digital-assets marketplace for music** *Next.js, TypeScript, Supabase, Vercel*

Built an end-to-end digital commerce platform, translating a real business workflow into software covering product presentation, checkout, payment state, data persistence, integrations, and post-purchase automation.

**RendeCerto — Investment-scenario simulator** *Next.js, TypeScript, Vitest*

Built a financial simulation and calculation application, translating investment scenarios into deterministic business rules and calculation logic, with automated testing to validate results across different scenarios.

**Quebrando Fronteiras — Social-impact and donation platform** *Next.js, TypeScript*
Refactored critical user flows in a social platform by identifying UX and flow problems and implementing improvements focused on clarity and conversion.

**Marca Racha — Local match discovery and coordination** *Next.js, React, Supabase, PostgreSQL, Prisma, Resend, Zod, Vitest*
Implemented group goals, invitations, and declared contributions, connecting authentication, persistence, and notifications. Behavioral tests cover query-boundary ownership, role-derived capabilities, membership-aware visibility, invitation limits, and state transitions.

**Saúde em Campo — Offline-first field-health operations** *React Native, Expo, TypeScript, SQLite, Drizzle, TanStack Query*
Designed a system to support field visits and health records when reliable connectivity is unavailable. The proposal defines per-user local data isolation, a sync queue, idempotency, retries/backoff, role-aware navigation, RBAC, and risk-prioritization rules; it is architecture evidence, not a clinical-deployment claim.

**Missões do Dia — Android application** *Java, SQLite, SharedPreferences, Services, BroadcastReceiver, AsyncTask, XML*
Built an academic mission-CRUD application with local persistence, native Android components, and XML layouts.

## EDUCATION

**Tecnólogo in Internet Systems** — UNCISAL (Brazil) *Completed 2026*

## LANGUAGES

**Portuguese:** Native ▪ **English:** Advanced